

Preservation by Species: Reframing Data Integrity Through Evolutionary Identity

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Abstract

Conventional approaches to digital preservation prioritize bit-level fidelity, treating any deviation from an original file state as corruption. This thesis proposes an alternative framework, drawing on evolutionary biology, where preserved works are understood as species concepts rather than immutable objects. Identity is defined by constrained recognisability, documented provenance, and structural intent, while individual encodings function as mutable carriers analogous to genetic material. Bounded variation enables continuity over time; preservation fails only when divergence exceeds established thresholds of recognition.

1. The Limits of Fidelity-Based Preservation

Digital preservation has traditionally been framed as a problem of accuracy, verified through checksums or cryptographic hashes. This assumes meaning is inseparable from precise technical configuration. However, these assumptions are fragile as formats become obsolete and storage media degrades. Furthermore, synthetic reproduction can now generate artifacts indistinguishable from originals without historical continuity. In such environments, a technically perfect copy without provenance may be less trustworthy than an imperfect artifact with a traceable transformation history.

2. Preservation as an Act of Recognition

Archivists routinely exercise judgment to determine if a migrated artifact still constitutes the same work. This framework formalizes that judgment, seeking to define the conditions under which these articulations of authenticity are defended.

3. The Misapplied DNA Analogy in Digital Discourse

While DNA is often cited as an ideal storage medium, biological systems do not actually preserve exact sequences. Mutation is a generative condition of persistence where viability, not accuracy, is defended. Identity persists through constrained variation rather than fixity.

4. File Identity as a Species Concept

Digital works should be viewed as species defined by shared characteristics and ancestry. The work is the species, individual files are organisms, and formats are the genetic material. Preservation shifts from enforcing exact replication to maintaining recognizable identity within declared bounds.

5. Constraint Rather Than Fidelity

What must be preserved is the “outline” of the work: its origin, structural grammar, and semantic intent. Surface-level fidelity is secondary to these core traits.

6. Error as Bounded Drift

Error is only destructive when it becomes unbounded. Loss occurs when independently reconstructed instances no longer converge on the same recognizable identity, rather than when a checksum simply breaks.

7. Lineage Over Replication

Species persist through lineage; preserved digital works persist through documented transformation histories. Each migration or modification is part of a work's evolution. In an age of trivial synthetic reproduction, provenance is the primary mechanism for defending authenticity.

8. Implications for AI and Synthetic Media

AI makes appearance insufficient as proof of origin. Authenticity must be decoupled from appearance and reattached to the legibility of an artifact's evolution.

9. Conclusion: Preserving Identity in Change

Digital preservation must adopt continuity through constrained transformation. We must preserve the conditions under which data can be recognized as originating from a specific source.

10. From Conceptual Model to Archival Practice

This shift requires managing variation over time and positioning lineage as the primary carrier of authenticity.

- **10.1 Defining the Species Descriptor:** Each work is associated with a profile specifying origin, structural grammar, semantic intent, and permissible variation.
- **10.2 Lineage Mechanisms:** Transformation paths (migrations, AI-assisted modifications) are recorded using frameworks like W3C PROV. An object remains authentic if its history is continuous and bounded.
- **10.3 Error as Signal:** Checksums become early-warning mechanisms rather than absolute arbiters. Success is defined by the convergence of transformed instances.
- **10.4 AI as Recorder:** AI systems support documentation by detecting transformations and evaluating semantic continuity, but they do not determine identity.
- **10.5 Scalability:** Costs shift from storage redundancy toward metadata intelligence and governance.